

Experiment No: 4

Title: Write a Python Program for the Crypt-Arithmetic Problem

Aim

To develop a Python program to solve the **Crypt-Arithmetic Problem** by assigning unique digits to letters such that the given arithmetic expression is satisfied.

Objective

To implement the Crypt-Arithmetic problem in Python by assigning unique digits to alphabets and verifying the correctness of the given arithmetic expression.

Algorithm

- **Step 1:** Start the program.
- **Step 2:** Read the words from the user.
- **Step 3:** Identify all unique letters in the words.
- **Step 4:** Generate possible digit assignments for the letters.
- **Step 5:** Assign a unique digit to each letter.
- **Step 6:** Check whether the arithmetic equation is satisfied.
- **Step 7:** If a valid assignment is found, display the solution.
- **Step 8:** Stop the program.

Program

```
from itertools import permutations

# Read input
word1 = input("Enter First Word: ").upper()
word2 = input("Enter Second Word: ").upper()
result = input("Enter Result Word: ").upper()

letters = list(set(word1 + word2 + result))

if len(letters) > 10:
    print("Too many unique letters.")
    exit()
```

```
digits = '0123456789'
```

```
found = False
```

```
for perm in permutations(digits, len(letters)):
```

```
    table = dict(zip(letters, perm))
```

```
    # Leading letter cannot be zero
```

```
    if table[word1[0]] == '0' or table[word2[0]] == '0' or table[result[0]] == '0':
```

```
        continue
```

```
    n1 = int("".join(table[ch] for ch in word1))
```

```
    n2 = int("".join(table[ch] for ch in word2))
```

```
    n3 = int("".join(table[ch] for ch in result))
```

```
    if n1 + n2 == n3:
```

```
        print("\nSolution Found\n")
```

```
        for k in sorted(table):
```

```
            print(k, "=", table[k])
```

```
    print("\n", n1)
```

```
    print("+", n2)
```

```
    print("-----")
```

```
    print(n3)
```

```
    found = True
```

```
    break
```

```
if not found:
```

```
    print("No Solution Found.")
```

Sample Input

Enter First Word: SEND

Enter Second Word: MORE

Enter Result Word: MONEY

Sample Output

```
Enter First Word: SEND
Enter Second Word: MORE
Enter Result Word: MONEY
```

```
Solution Found
```

```
D = 7
E = 5
M = 1
N = 6
O = 0
R = 8
S = 9
Y = 2
```

```
  9567
+ 1085
-----
 10652
```

Result

Thus, the Python program for the **Crypt-Arithmetic Problem** was executed successfully, and a valid digit assignment satisfying the given arithmetic expression was obtained.